Lab: Blender Lighting & Cameras

Let's imagine we were filming a movie or live tv show. In additional to the actors, actresses, props, and sets we need to light the scene. Traditional film studios have light rigs that apply lighting to the set and light up the actors and actresses. They also would need a camera to record all the action. This camera is the window into the world the director has created and is what the audience will eventually see.

Blender allows us to create a virtual world with virtual objects and apply materials to them. Blender also allows us to light our virtual world using Blenders lighting system.

This lab will go over the basics of lighting and recording / rendering the scene using a camera.

Setup



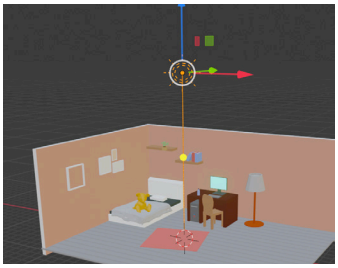
Open the file light-cameras.blend. The file contains a basic interior setup of a bedroom. The setup almost appears as if it's a set up in a movie studio.

Add Light to Our Scene



Lighting is a complex subject, there are people in Hollywood who specialize in how to use lights effectively in a film production. This lab will just cover the basics.

Switch the scene to *Display Render Preview* or hit the '*z*' key and select *Rendered*. This option shows your scene with lighting be applied. At the moment our scene is dark because there are no lights in our scene. Think about it a real world setting, if it is nighttime and there are no lights on in your home, how would you see anything?

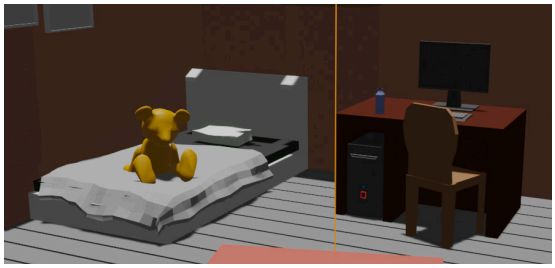


Let's begin by adding some base light to our scene. In Blender go to *Add > Light > Sun*, this will add a dashed star in the center of the scene. The sun object add light to ALL objects in your scene, just like how the sun adds light to all objects on Earth.

Put the sun object into the *Lights & Camera* collection in the outliner. You can then use the move, scale and rotate tools just like we have with 3D meshes. Move the sun so it above the scene, roughly in the middle.

Click on the sun object and make sure you are in object mode. Go to the properties panel to change the properties of the light. The main property we will affect is the ***strength***. The strength is how strong is the light in our scene, it goes 0 - 10. Let's set it to a strength of *3*. To see the effect of the sun light to our scene preview the scene in '*Rendered*' mode. It should look roughly like the scene on the right.

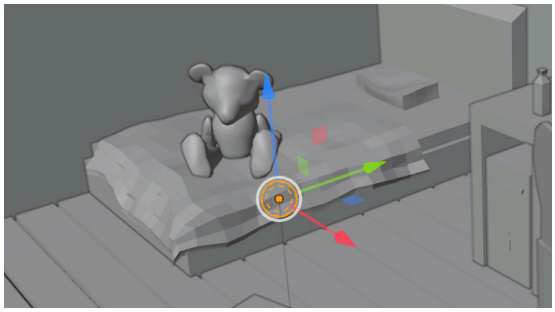
Go back and set the sun's strength level to 6 and view in '*Rendered'* again, you should see the scene get brighter.



Lights in our scene not only make the scene brighter, but they objects in the scene cast shadows. We can turn off shadows if we like by unclicking the '*Shadow*' option.

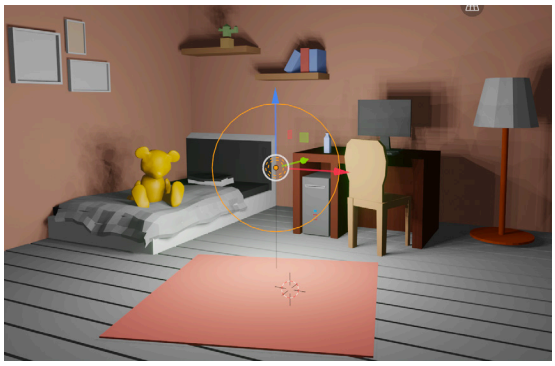
Add a Point Light

Now that we have a base light let's add some more lights to our scene to give it some more detail. Let's create a *Point Light* for the scene.



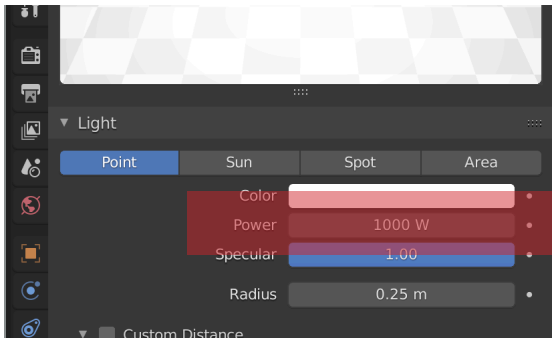
Unlike the sun, the point light does not affect all objects, but only objects in the nearby area. The point light release light in all direction, similar to a light bulb in the real world.

Place your point light into the *Lights & Camera* collection. Move the area light to the center of the room. Unlike the sun light object, the position of the point light directly affects the lighting in the scene.



Turn off the sun object so that you point light is the only light in the scene. Render the scene again and you'll see that the scene is darker than when we were using the sun object. The point light at the moment is weak and not affecting all the objects in the scene, so they are dark.

The strength of your point light is determined by the *Power* option. We measure the strength of light as wattage. The higher the wattage the stronger the light source.



Set your scene to *500* watts and render it. You'll see that the light source is the center of the room, the objects further away from the light are less bright. Raise the power level from *500w* to *1000w*, you'll see that the room is getting brighter. Set it again to *2000w* and the room is starting to filled with light.

Add an Area Light



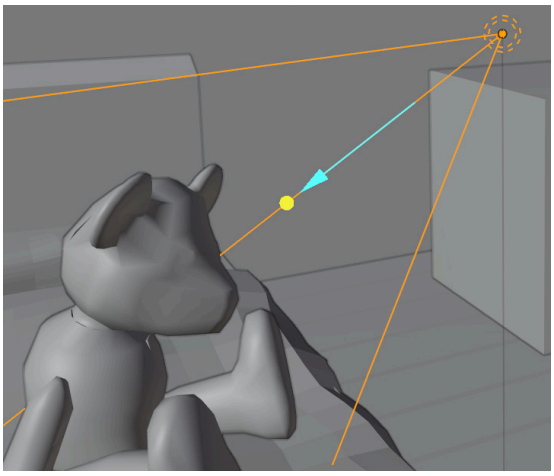
We are using the point light to light our scene, but we will sometimes want additional light sources. Take a look at the lamp at the far right corner. If the lamp were on it would release lighting down towards the floor. To do this we will add an area light.

Go to *Object > Add > Area Light*. Use the move tool to move the area light inside of the lamp shade. Use the rotate tool to point the area light down towards the floor.



Set the wattage to *1000w*. View the scene and you'll see that there are two light sources affecting the scene. To see the difference you can turn on and off lights using the outliner in Blender.

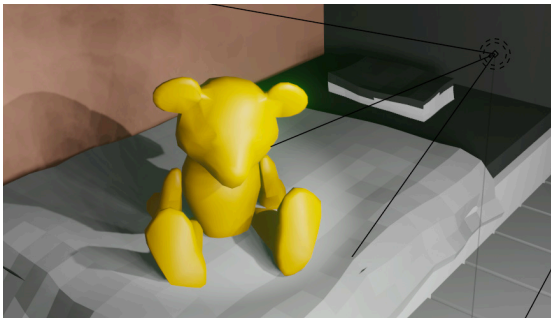
Add a Spot Light



You may have heard of spotlights. They are used in theaters and film sets to add light to a specific spot within a scene. Typically spot lights are used light to the main character within the scene.

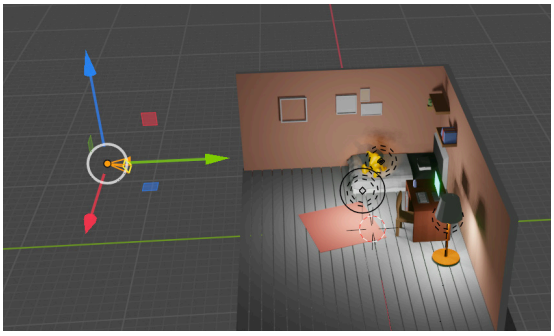
In this example, let's add a spotlight to the teddy bear on the bed, this will give the teddy bear emphasis.

Create a spot light by going to *Add > Light > Spot*. Use the move tool to position the spot light above the bed / teddy bear. Unlike the point light the spot light only focuses light in one direction. Use the rotate option to point the cyan line towards the teddy bear.



Like the area and point light increase the wattage to increase the intensity of the light. Set the wattage to 500 watts. Go to rendered mode you'll see the light shines a hard edge along the bed and wall. To soften the edge of the spotlight go to *Spot Shape > Blend* and increase the blend amount. The higher the blend amount the softer the edges of the light will be.

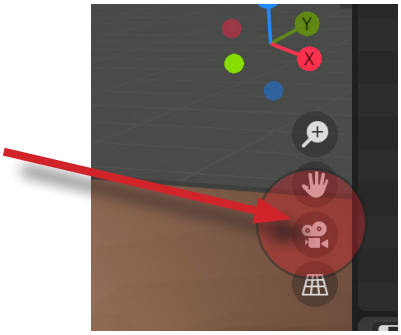
Cameras



Now that we our scene set up and lit, let's take a photo! In order to record or export a photo of our scene in Blender we need to build a camera. The camera we build into our scene will be the window into the world. When we export / render a photo it will be from this camera, if we were exporting a 3D animation from Blender it would be from the vantage point of the camera.

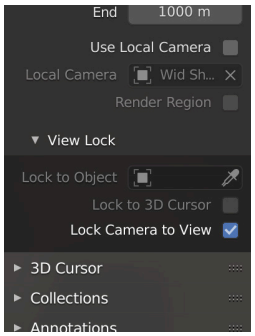
To build a camera in Blender go to *Add > Camera*. This will build a camera object into the center of your scene. Put your camera into the *Lights & Cameras* collection in the outliner. Rename it

'Wide Shot Camera'.



To move and position the camera in Blender use the object mode move and rotate tools. Place the camera above the scene looking down towards the teddy bear at an angle.

To view the vantage from the camera click the camera icon in the upper right hand corner of the 3D Viewport. You can also go to *View > Cameras > Active Camera*. This will give you the camera view of the scene. The actual shot is outlined in yellow, the area grayed out will not recorded. Click the camera again to leave the camera view or simply orbit around the scene.

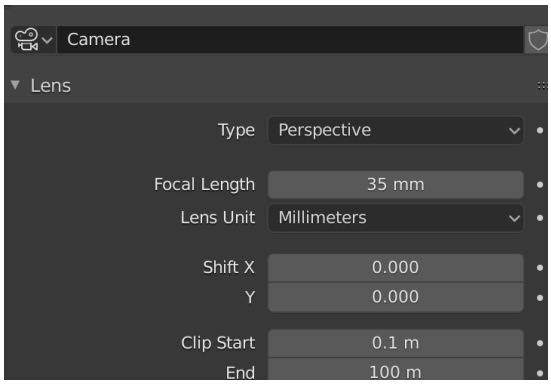


Another way to move the camera is open the *Sidebar* in Blender by going to *View > Sidebar* or use the *'n'* button hotkey. Go to *View > Lock Camera* to View and turn on the checkbox. This will lock you camera to our world view, as you orbit, pan and zoom, so will the camera.

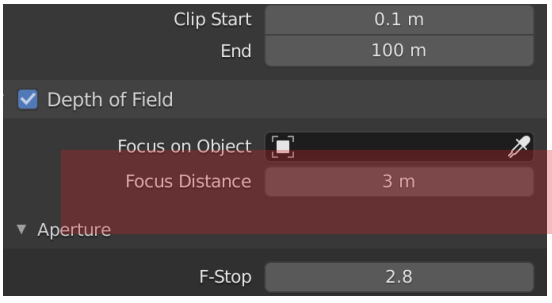
Camera Settings

Now that we have built and positioned our camera let's take a look at the different settings we can apply. Cameras in the real world have lens, these lens determine the focal distance of the camera. For instance, a 35mm lens typically makes the scene look zoomed out and far. A higher mm lens such as 50mm lens will make the same camera and the same position appear closer and tighter.

Let's apply the same thing in our scene. Select the camera and choose the camera icon in the properties panel. Go under the lens properties and change the *Focal Length*. Change it from *35mm* to *50mm*. View your camera again and the scene will seem a bit tighter than before. Change the focal length to *85mm*, this will make the image seemed even more zoomed.



Another setting we can tinker with is the *Depth of Field*. In the real world cameras have something called a depth-of-field, which is their ability to focus on objects in an environment. Typically cameras can not focus on everything in the scene if the scene contains a lot of information. If the camera focuses on the foreground than the background will appear out of focus, or if the camera will focus on the background and the foreground will appear out of focus. Depth-of-field is photography term for how much of the scene will appear in focus.

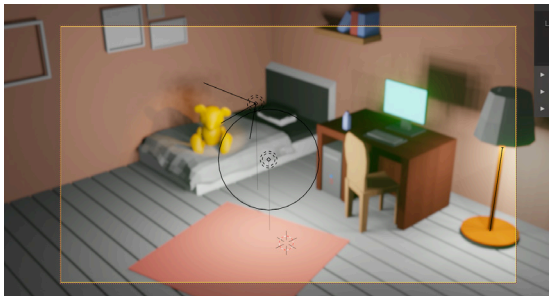


In Blender the *Depth of Field* option allows us to set the focal distance. Objects inside of the focal distance will appear in focus, and those farther than the distance will appear out of focus.

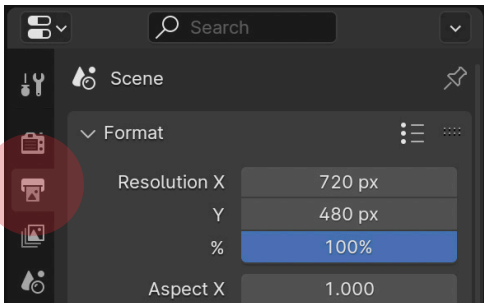
Rendering



In order to see the depth-of-field we will need to render our image. Rendering is when Blender takes all of the information in the scene including the meshes, materials, lighting, cameras and more and produces an image or movie. Depending on the complexity of your scene and the hardware / graphics card in your computer the rendering may take a few seconds or more.



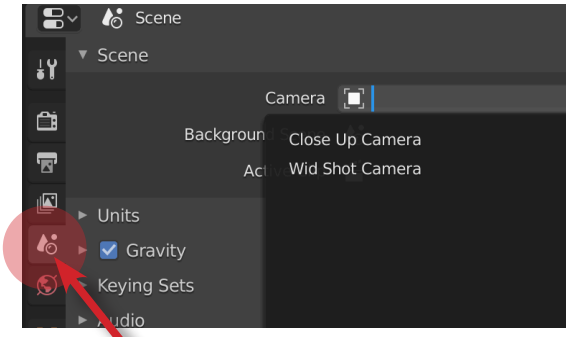
To render your scene position the camera in your scene look down on the set from above, but capture the whole room. In the *Depth of Field* set the focus distance to 1m. *Go to Render > Render Image*. Your scene should render it will look blurry since the focal distance is so short. Let's change the focal distance from *1* to *3*. Render the scene again and you'll see that the scene is a bit more in focus. Set the *Focal Distance* to *10* and render the image and it should appear in full focus.



To set the dimensions of your image click on the Output tab in the Properties Panel, it will look like an old school printer.

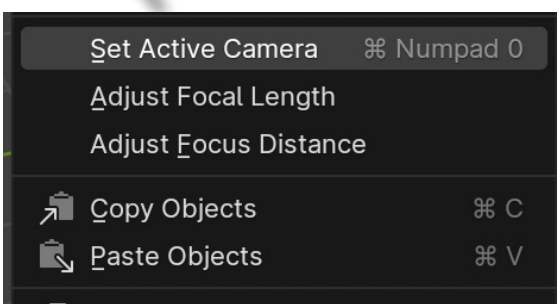
Under resolution you will see it is defaultly set to HD or *1920px x 1080px.* Let's make it smalle and set it to *720px x 480px*. Go to *Render > Render Image* and you will see that the new image is smaller than the previous one.

Create a Second Camera



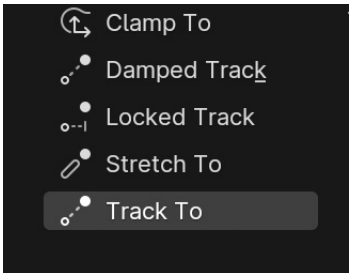
In many Hollywood movie multiple cameras are used to film the scene from different angles. For this Blender scene we'll create a second camera to focus on our scene's star, the teddy bear!

Go to *Add > Camera* to create a second camera to our scene. Make sure to add it to *Lights & Cameras* collection in the outliner. Name the camera, *Close Up Camera*. Position the camera as you did with the previous camera.



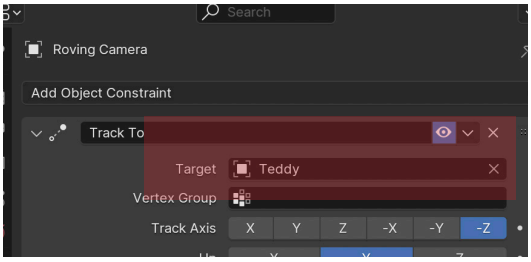
Position the new camera to focus on the teddy bear's face. Blender can only have one active camera to render a scene with, thus, we may want to switch between different cameras that we have created. To switch cameras click on the *Scene Properties* option in the properties panel. Under the camera switch the camera from the *Wide Shot Camera* to the *Close Up Camera*. Render a new image and you should that the new camera perspective is shown. Another way to switch the active camera is to right click on the camera and choose Set Active Camera from the pop up menu.

Add a camera / light with a contraint

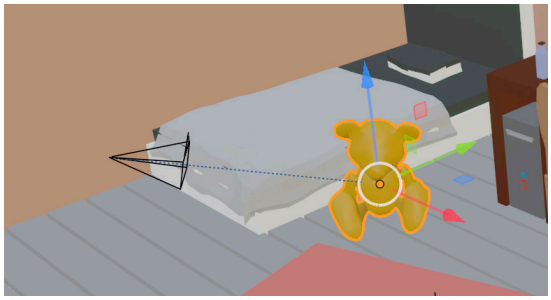


Sometimes it would be nice to have our camera or light follow the character, rather than needing to re-adjust them if we decide to move the character. We can do this using something called a constraint. Like it sounds like it will constrain the way the object can move.

Go to *Add > Camera* to add a third camera to the scene and call this *Roving Camera*. Use the move / rotation tool to have the camera pointing at an angle at the teddy bear. With the *Roving Camera* selected go to the *Constraints* menu, it has a blue icon. Click on the *Add Constrainst*s button and choose *Track To*. Like it sounds, this means the object, in this case the camera will track and object.

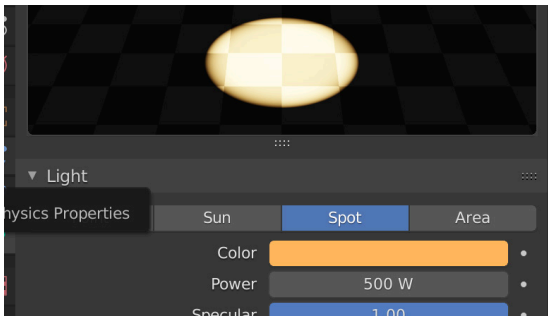


Under *Target* choose *Teddy Bear.* This means that the *Roving Camera* will also track or point towards the teddy bear. Select the teddy bear and move it around and you will see that the Roving Camera automatically rotates.



Add Mood Lighting

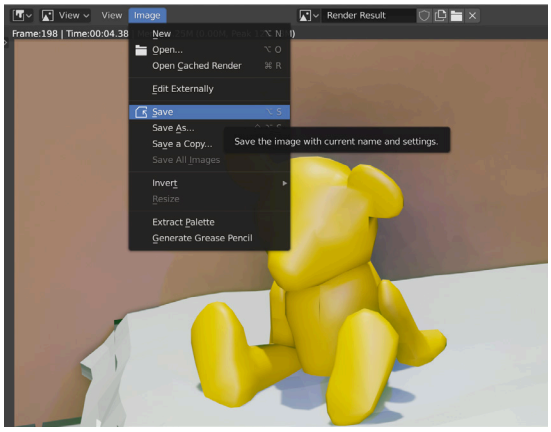
Color lights in a scene can help affect the mood of a scene. Warm colors like red, orange and blue can help create a sense of warmth for a character. Cool colors like blue, green and purple can make a scene feel more subdued. In the real world colored light bulbs and light gels are used to create this affect, we can also create this affect in Blender as well.



Click on your spotlight that created that focuses on the teddy bear. Under the light properties change the *Color* property of the light from white to a medium orange. Look at the scene and the teddy bear feels a bit warm-er.



Render / Save an Image



Let's put all of this together. Let's save a shot of the teddy bear with the orange glow lighting.

Select your *Close Up Camera* and set that to the active camera. Go to *Render > Render Image*. After rendering an image, go to *Image > Save* and save the render image to your computer as *warm-light.png*.

Go back and change the lighting for the spotlight to cool lighting. Set the spot light to a cool color such as blue or purple. Render the scene again and save the image as *cool-light.png*.

Hopefully this lab has helped you understand for lighting and camera work can effect your Blender scene.

Submission

Submit a folder containing the Blender file as well as two rendered images.